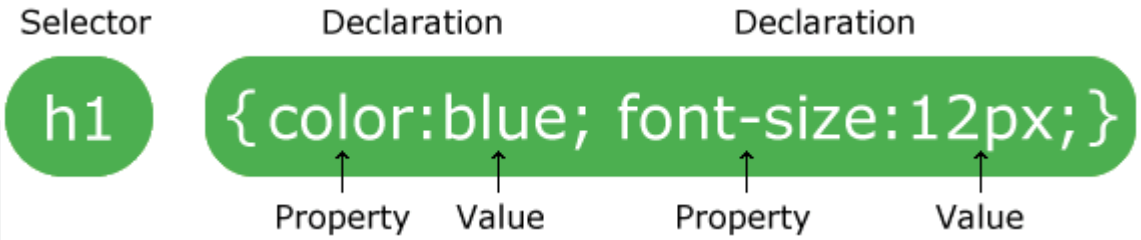
Introduction to CSS

- **CSS** stands for **c**ascading **s**tyle **s**heets.
- CSS describes **how HTML elements are to be displayed on screen or in other media**
- CSS **saves a lot of work** as it can control the layout of multiple web pages all at once.

CSS SYNTAX



- The declaration block contains one or more declarations separated by semicolons.
- Each declaration includes a CSS **Property Name** and a **Value**, separated by a colon.
- A CSS declaration always ends with a semicolon, and declaration blocks are surrounded by curly braces.

Run ❯

Result Size: 753 x 512

Get your own website

```
<!DOCTYPE html>
<html>
<head>
<style>
body {
  background-color: lightblue;
}

h1 {
  color: white;
  text-align: center;
}

p {
  font-family: verdana;
  font-size: 20px;
}
</style>
</head>
<body>

<h1>P.P Savani University</h1>
<p>School of Engineering</p>

</body>
</html>
```

P.P Savani University

School of Engineering

Types of CSS : Inline Styles

- An inline style may be used to apply a unique style for a single element. (**It is apply for Single Line)**.
- To use inline styles, add the **style attribute** to the relevant element.
- The style attribute can contain any CSS property.
- **Example:**

```
<h1 style="color : blue;"> PU</h1>
```

Internal Styles

- **An internal style sheet used for one single page for a unique style.**
- Internal styles are defined within the **<style> element**, **inside the <head>** section of an HTML page:
- **Example:**

```
<head>
    <style>
        body {background-color: linen;}
        h1{color: maroon;}
    </style>
</head>
```

External Styles

- With an external style sheet, you can change the look of an entire website by changing just one file. **(This style is used for multiple web pages)**
- Each page must include a **reference to the external style** sheet file inside **the <link> element**. The <link> element goes inside the <head> section:

```
<head>
<link rel="stylesheet" type="text/css" href="mystyle.css">
</head>
```

Mystyle.css

```
body { background-color: blue;}
```

CSS TEXT & PROPERTIES

- **Color**
- **Text-align**
- **Text-decoration**
- **Text-transform**
- **Text-indent**
- **Text-shadow**
- **Letter-spacing**
- **Line-height**

```
P
{
Color:blue;
Text-align:right;
Text-decoration:underline;
Text-transform:lowercase;
Text-indent:10px;
Text-shadow:2px 2px 2px #ff1234;
Letter-spacing:2px;
Line-height:60%
}
```

CSS FONTS

- **Font**
- **Font-family**
- **Font-size**
- **Font-style**
- **Font-variant**
- **Font-weight**

```
P
{
Font:italic bold 12px serif;
Font-family:”Times New
Roman”,”serif”;
Font-size:small;
Font-style:normal;
Font-variant:normal;
Font-weight:bold;
}
```

CSS LINK

- **Unvisited Link**

  a:link {color: #123456;}

- **Visited Link**

  a:visited {color:#FF0500;}

- **Mouse Over Link**

  a:hover { color:#1234FF;}

- **Selected Link**

  a:active { color:#FF10FF;}

CSS BACKGROUND

- **Background**: write color or url of image
- **Background-attachment**: attachment is fixed or not
- **Background-color**: color of background
- **Background-image**: url of image
- **Background-position**: center, left, right
- **Background-repeat**: repeat or not repeat

CSS Selector

- The **element selector** selects **HTML elements based on the element name.**

```
<!DOCTYPE html>
<html>
<head>
<style>
p {
  text-align: center;
  color: red;
}
</style>
</head>
<body>

<p>Every paragraph will be affected by the style.</p>
<p id="para1">Me too!</p>
<p>And me!</p>

</body>
</html>
```

Every paragraph will be affected by the style.

Me too!

And me!

CSS id Selector

- The id selector uses the id attribute of an HTML element to **select a specific element**.

- The id of an element is **unique within a page, so the id selector is used to select one unique element**.

- To select an element with a **specific id, write a hash (#) character, followed by the id of the element**.

```html
<!DOCTYPE html>
<html>
<head>
<style>
#para1 {
  text-align: center;
  color: red;
}
#para2 {
  color: blue;
}
</style>
</head>
<body>

<p id="para1">Hello World!</p>
<p id="para2">This paragraph is not affected by the style.</p>

</body>
</html>
```

Hello World!

This paragraph is not affected by the style.

CSS class Selector

- The class selector selects HTML elements with a specific class attribute.

- To select elements with a specific class**, write a period (.) character, followed by the class name.**

```html
<!DOCTYPE html>
<html>
<head>
<style>
.center {
  text-align: center;
  color: red;
}
</style>
</head>
<body>

<h1 class="center">PPSU</h1>
<p class="center">SOE</p>

</body>
</html>
```

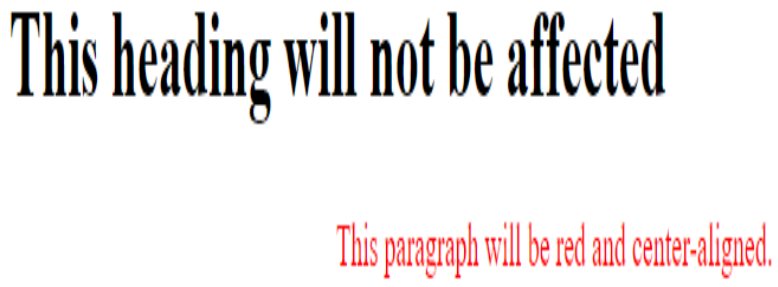
PPSU

SOE

```html
<!DOCTYPE html>
<html>
<head>
<style>
p.center {
  text-align: center;
  color: red;
}
</style>
</head>
<body>

<h1 class="center">This heading will not be affected</h1>
<p class="center">This paragraph will be red and center-aligned.</p>

</body>
</html>
```

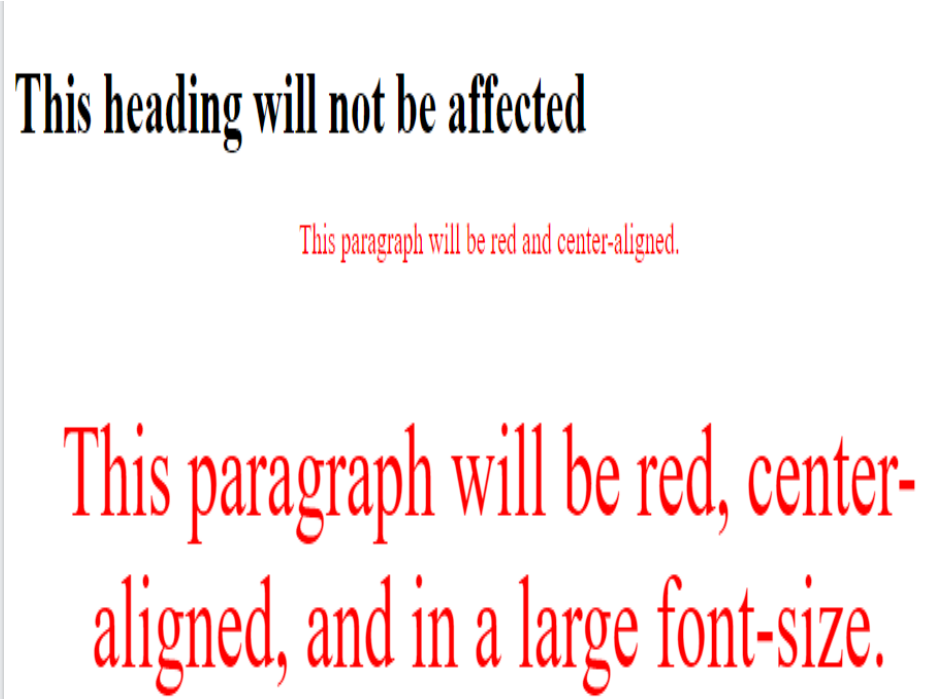


```
<!DOCTYPE html>
<html>
<head>
<style>
p.center {
  text-align: center;
  color: red;
}

p.large {
  font-size: 300%;
}
</style>
</head>
<body>

<h1 class="center">This heading will not be affected</h1>
<p class="center">This paragraph will be red and center-aligned.</p>
<p class="center large">This paragraph will be red, center-aligned, and in a large font-size.</p>

</body>
</html>
```



CSS Universal Selector

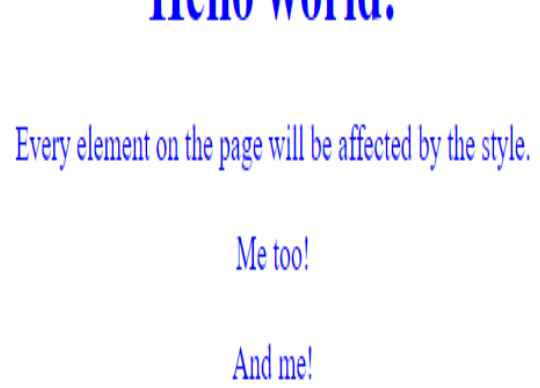
- The universal selector (*) selects all HTML elements on the page.

```
<!DOCTYPE html>
<html>
<head>
<style>
* {
  text-align: center;
  color: blue;
}
</style>
</head>
<body>

<h1>Hello world!</h1>

<p>Every element on the page will be affected by the style.</p>
<p id="para1">Me too!</p>
<p>And me!</p>

</body>
</html>
```



CSS Grouping Selector

- The grouping selector selects all the HTML elements with the same style definitions.

- Look at the following CSS code (the h1, h2, and p elements have the same style definitions)

```html
<!DOCTYPE html>
<html>
<head>
<style>
h1, h2, p {
  text-align: center;
  color: red;
}
</style>
</head>
<body>

<h1>Hello World!</h1>
<h2>Smaller heading!</h2>
<p>This is a paragraph.</p>

</body>
</html>
```



Tag

- The tag defines a **division or a section** in an HTML document.

- The tag is used as a container for HTML elements - which is then styled with CSS

```
<!DOCTYPE html>
<html>
<head>
<style>
.myDiv {
  border: 5px solid red;
  background-color: lightpink;
  text-align: center;
}
</style>
</head>
<body>

<h1>The div element</h1>

<div class="myDiv">
  <h2>This is a heading in a div element</h2>
  <p>This is some text in a div element.</p>
</div>

<p>This is some text outside the div element.</p>

</body>
</html>
```

The div element

This is a heading in a div element

This is some text in a div element.

This is some text outside the div element.

CSS Padding

- Padding is used to **create space around an element's content, inside of any defined borders**.
- With CSS, you have full control over the padding. There are properties for setting the padding for each side of an element (**top, right, bottom, and left**).



- CSS has properties for specifying the padding for each side of an element:
  - padding-top
  - padding-right
  - padding-bottom
  - padding-left
- All the padding properties can have the following values:
  - length - specifies a padding in **px, pt, cm, etc**.
  - % **- specifies a padding in % of the width of the containing element**

```html
<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid black;
  background-color: yellow;
  padding-top: 50px;
  padding-right: 30px;
  padding-bottom: 50px;
  padding-left: 80px;
}
</style>
</head>
<body>

<h2>PPSU</h2>

<div>This div element has a top padding of 50px, a right padding of 30px, a bottom padding of 50px, and a left padding of 80px.</div>

</body>
</html>
```

PPSU

This div element has a top padding of 50px, a right padding of 30px, a bottom padding of 50px, and a left padding of 80px.

Padding - Shorthand Property

- The padding property is a shorthand property for the following individual padding properties:

- padding-top
- padding-right
- padding-bottom
- padding-left

- If the padding property has **four values**:

- **padding: 25px 50px 75px 100px;**
- top padding is 25px
- right padding is 50px
- bottom padding is 75px
- left padding is 100px

```html
<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid black;
  padding: 25px 50px 75px 100px;
  background-color: lightblue;
}
</style>
</head>
<body>

<h2>The padding shorthand property - 4 values</h2>

<div>This div element has a top padding of 25px, a right padding of 50px, a bottom padding of 75px, and a left padding of 100px.</div>

</body>
</html>
```

The padding shorthand property - 4 values

This div element has a top padding of 25px, a right padding of 50px, a bottom padding of 75px, and a left padding of 100px.

- If the padding property has three values:

- **padding: 25px 50px 75px;**

- **top padding** is 25px
- **right and left paddings** are 50px
- **bottom padding** is 75px

```html
<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid black;
  padding: 25px 50px 75px;
  background-color: lightblue;
}
</style>
</head>
<body>

<h2>The padding shorthand property - 3 values</h2>

<div>This div element has a top padding of 25px, a right and left padding of 50px, and a bottom padding of 75px.</div>

</body>
</html>
```

The padding shorthand property - 3 values

This div element has a top padding of 25px, a right and left padding of 50px, and a bottom padding of 75px.

- If the padding property has two values:

- padding: 25px 50px;
- **top and bottom paddings** are 25px
- **right and left paddings** are 50px

```html
<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid black;
  padding: 25px 50px;
  background-color: lightblue;
}
</style>
</head>
<body>

<h2>The padding shorthand property - 2 values</h2>

<div>This div element has a top and bottom padding of 25px, and a right and left padding of 50px.</div>

</body>
</html>
```

The padding shorthand property - 2 values

This div element has a top and bottom padding of 25px, and a right and left padding of 50px.

- If the padding property has one value:

- **padding: 25px;**
- all four paddings are 25px

```html
<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid black;
  padding: 25px;
  background-color: lightblue;
}
</style>
</head>
<body>

<h2>The padding shorthand property - 1 value</h2>

<div>This div element has a top, bottom, left, and right padding of 25px.</div>

</body>
</html>
```

The padding shorthand property - 1 value

This div element has a top, bottom, left, and right padding of 25px.

Padding and element width

```
<!DOCTYPE html>
<html>
<head>
<style>
div.ex1 {
  width: 300px;
  background-color: yellow;
}

div.ex2 {
  width: 300px;
  padding: 25px;
  background-color: lightpink;
}
</style>
</head>
<body>

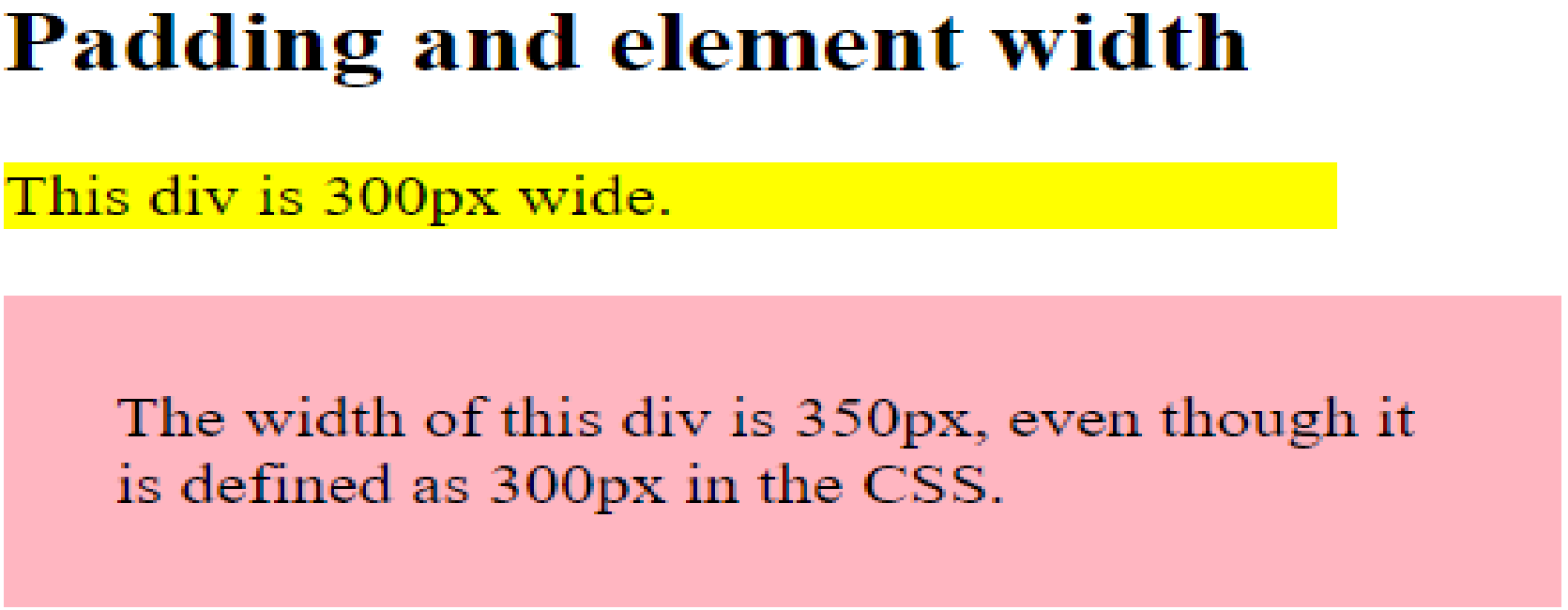
<h2>Padding and element width</h2>

<div class="ex1">This div is 300px wide.</div>
<br>

<div class="ex2">The width of this div is 350px, even though it is defined as 300px in
the CSS.</div>

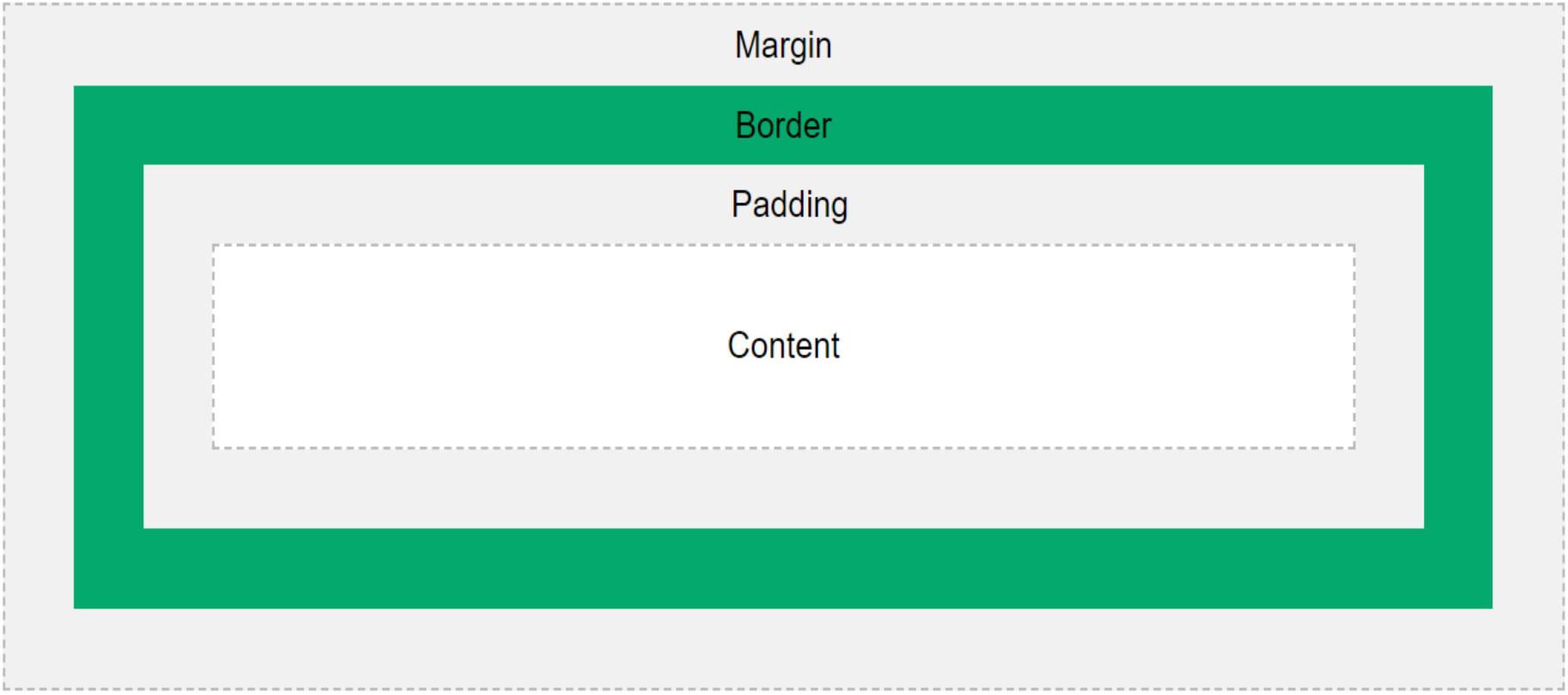
</body>
</html>
```

- element is given a width of 300px. However, the actual width of the element will be 350px (300px + 25px of left padding + 25px of right padding):



CSS Box Model

The CSS box model is essentially a box that wraps around every HTML element. It consists of: margins, borders, padding, and the actual content. The image below illustrates the box model:



Explanation of the different parts:

- **Content** - The content of the box, where text and images appear
- **Padding** - Clears an area around the content. The padding is transparent
- **Border** - A border that goes around the padding and content
- **Margin** - Clears an area outside the border. The margin is transparent

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  background-color: lightpink;
  width: 500px;
  border: 15px dotted red;
  padding: 40px;
  margin: 30px;
}
</style>
</head>
<body>

<h2>Box Model</h2>

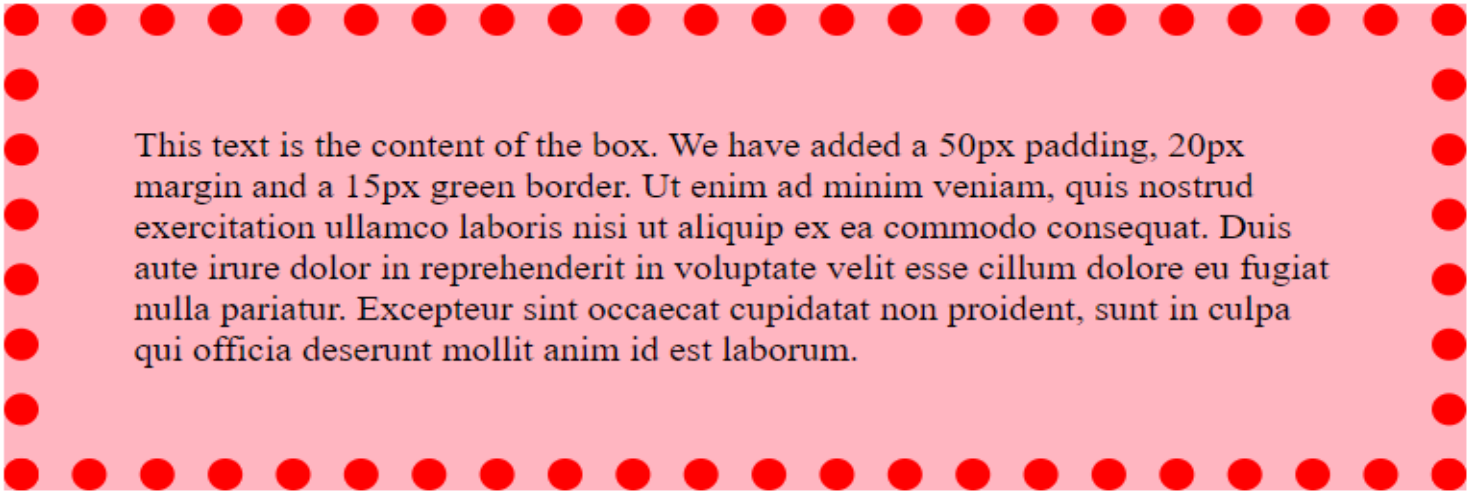
<p>The CSS box model is essentially a box that wraps around every HTML element. It consists of: borders, padding, margins, and the actual content.</p>

<div>This text is the content of the box. We have added a 50px padding, 20px margin and a 15px green border. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.
</div>

</body>
</html>
```

Box Model

The CSS box model is essentially a box that wraps around every HTML element. It consists of: borders, padding, margins, and the actual content.



```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  width: 320px;
  padding: 10px;
  border: 5px solid gray;
  margin: 0;
}
</style>
</head>
<body>

<h2>Calculate the total width:</h2>

<img src="klematis4_big.jpg" width="350" alt="Klematis">
<div>The picture above is 350px wide. The total width of this element is also 350px.
</div>

</body>
</html>
```

Calculate the total width:



The picture above is 350px wide. The total width of this element is also 350px.

Exercise

- Create a HTML Page which include text, image and heading with box model

CSS text-shadow Property

CSS Syntax

```
text-shadow: h-shadow v-shadow blur-radius color|none|initial|inherit;
```

h-shadow	Required. The position of the horizontal shadow. Negative values are allowed
v-shadow	Required. The position of the vertical shadow. Negative values are allowed
blur-radius	Optional. The blur radius. Default value is 0
color	Optional. The color of the shadow.
none	Default value. No shadow
initial	Sets this property to its default value.
inherit	Inherits this property from its parent element.

```
<!DOCTYPE html>
<html>
<head>
<style>
h1 {
   text-shadow: 2px 2px #FF0000;
}
</style>
</head>
<body>

<h1>The text-shadow Property</h1>

</body>
</html>
```

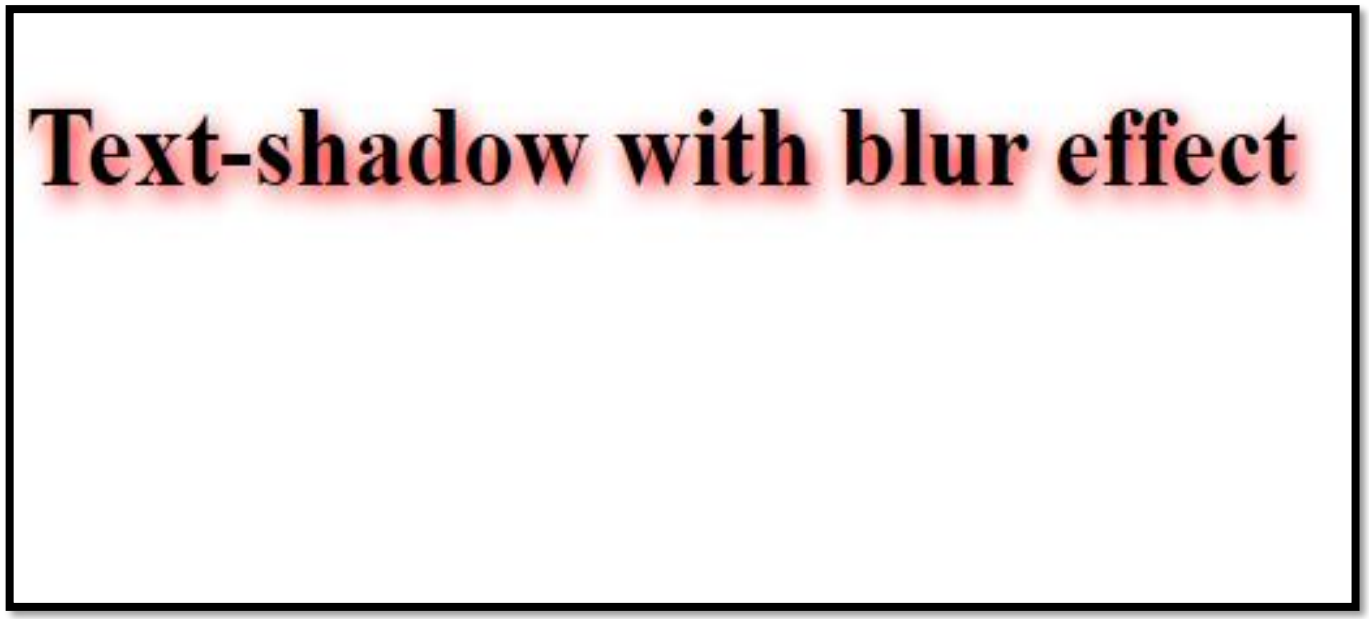
The text-shadow Property

Text-shadow with a blur effect

```
<!DOCTYPE html>
<html>
<head>
<style>
h1 {
   text-shadow: 2px 2px 8px #FF0000;
}
</style>
</head>
<body>

<h1>Text-shadow with blur effect</h1>

</body>
</html>
```


Text-shadow with blur effect

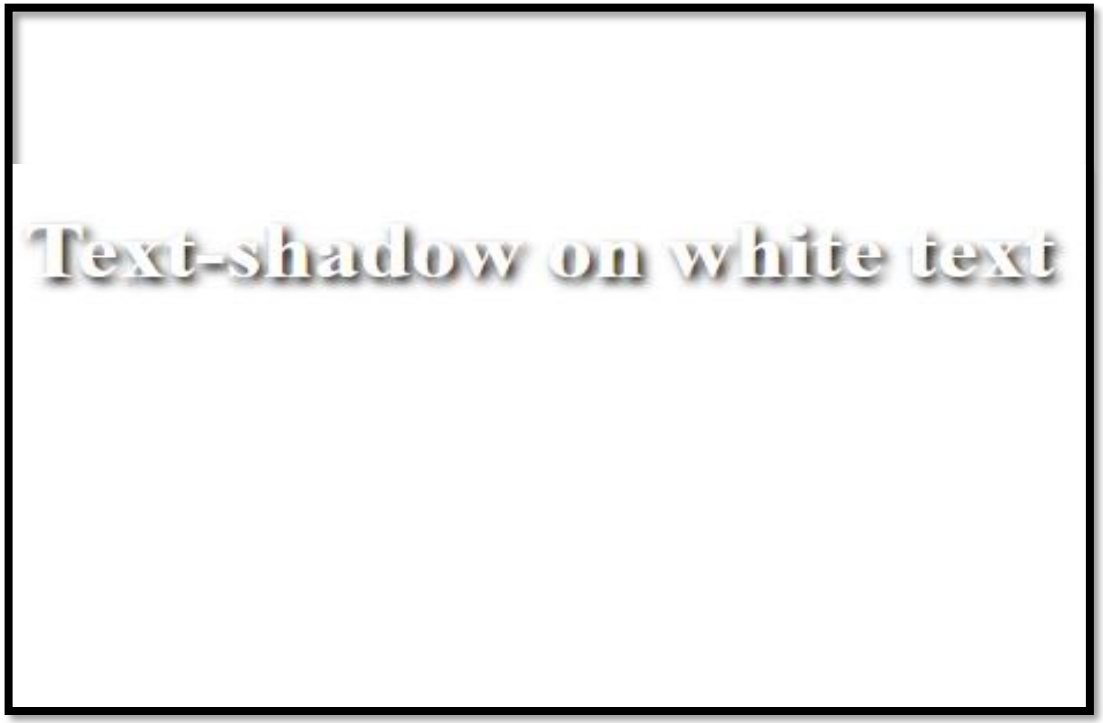
Text-shadow on a white text

```
<!DOCTYPE html>
<html>
<head>
<style>
h1 {
  color: white;
  text-shadow: 2px 2px 4px #000000;
}
</style>
</head>
<body>

<h1>Text-shadow on white text</h1>

</body>
</html>
```

Text-shadow on a white text



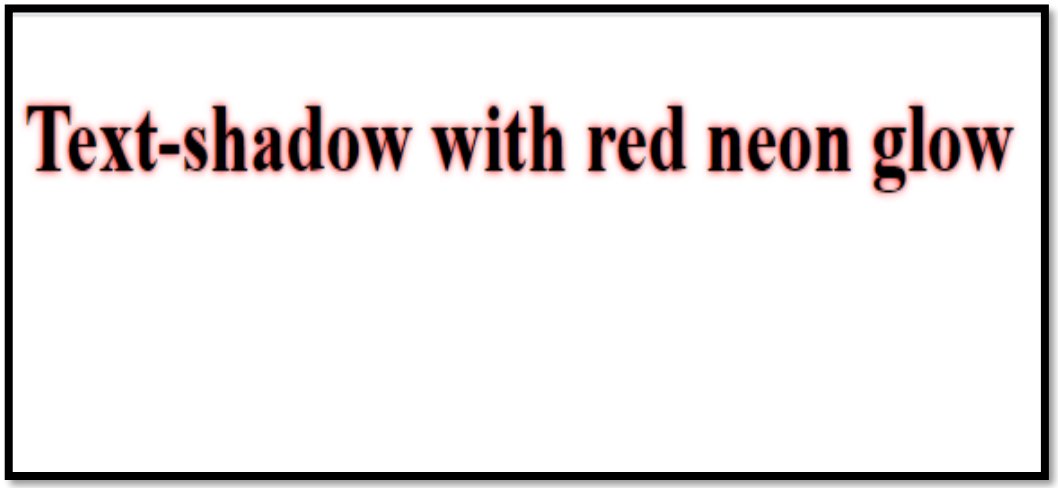
Text-shadow with a red neon glow

```
<!DOCTYPE html>
<html>
<head>
<style>
h1 {
  text-shadow: 0 0 3px #FF0000;
}
</style>
</head>
<body>

<h1>Text-shadow with red neon glow</h1>

</body>
</html>
```

Text-shadow with a red neon glow



Text-shadow with a red and blue neon glow

```
<!DOCTYPE html>
<html>
<head>
<style>
h1 {
  text-shadow: 0 0 3px #FF0000, 0 0 5px #0000FF;
}
</style>
</head>
<body>

<h1>Text-shadow with red and blue neon glow</h1>

</body>
</html>
```

Text-shadow with a red and blue neon glow



CSS Opacity / Transparency

- The opacity property specifies the opacity/transparency of an element.

Transparent Image

The `opacity` property can take a value from 0.0 - 1.0. The lower the value, the more transparent:



opacity 0.2



opacity 0.5



opacity 1
(default)

```html
<!DOCTYPE html>
<html>
<head>
<style>
img {
  opacity: 0.5;
}
</style>
</head>
<body>

<h1>Image Transparency</h1>
<p>The opacity property specifies the transparency of an element. The lower the value, the more transparent:</p>

<p>Image with 50% opacity:</p>
<img src="img_forest.jpg" alt="Forest" width="170" height="100">

</body>
</html>
```

Image Transparency

The opacity property specifies the transparency of an element. The lower the value, the more transparent:

Image with 50% opacity:



```html
<!DOCTYPE html>
<html>
<head>
<style>
img {
  opacity: 0.5;
}

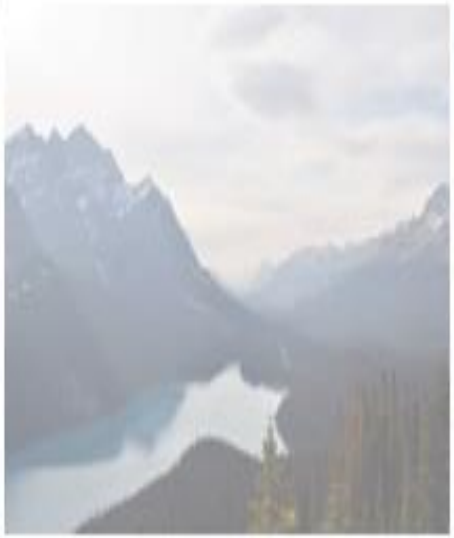
img:hover {
  opacity: 1.0;
}
</style>
</head>
<body>

<h1>Image Transparency</h1>
<p>The opacity property is often used together with the :hover selector to change the opacity on mouse-over:</p>
<img src="img_forest.jpg" alt="Forest" width="170" height="100">
<img src="img_mountains.jpg" alt="Mountains" width="170" height="100">
<img src="img_5terre.jpg" alt="Italy" width="170" height="100">

</body>
</html>
```

Image Transparency

The opacity property is often used together with the :hover selector to change the opacity on mouse-over:


```
<!DOCTYPE html>
<html>
<head>
<style>
img:hover {
  opacity: 0.5;
}
</style>
</head>
<body>

<h1>Image Transparency</h1>
<p>The opacity property is often used together with the :hover selector to change the opacity on mouse-over:</p>
<img src="img_forest.jpg" alt="Forest" width="170" height="100">
<img src="img_mountains.jpg" alt="Mountains" width="170" height="100">
<img src="img_5terre.jpg" alt="Italy" width="170" height="100">

</body>
</html>
```

Image Transparency

The opacity property is often used together with the :hover selector to change the opacity on mouse-over:





CSS Styling Links

```
<!DOCTYPE html>
<html>
<head>
<style>
a:link, a:visited {
  background-color: pink;
  color: white;
  padding: 14px 25px;
  text-align: center;
  text-decoration: none;
  display: inline-block;
}

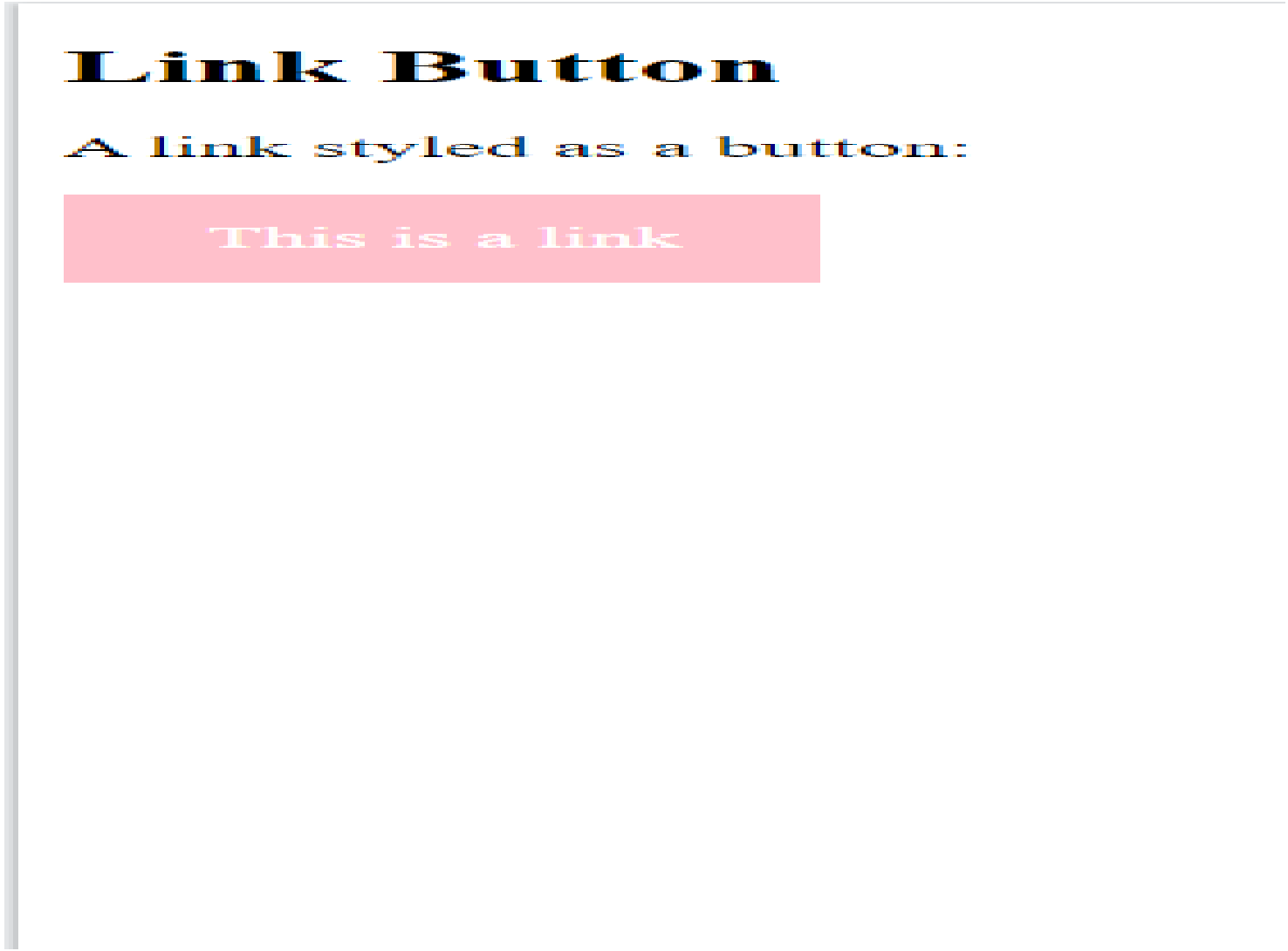
a:hover, a:active {
  background-color: red;
}
</style>
</head>
<body>

<h2>Link Button</h2>

<p>A link styled as a button:</p>
<a href="default.asp" target="_blank">This is a link</a>

</body>
</html>
```

CSS Styling Links



```
<!DOCTYPE html>
<html>
<head>
<style>
a:link {
  background-color: yellow;
}

a:visited {
  background-color: pink;
}

a:hover {
  background-color: lightgreen;
}

a:active {
  background-color: hotpink;
}
</style>
</head>
<body>

<h2>Styling a link with background-color property</h2>

<p><b><a href="default.asp" target="_blank">This is a link</a></b></p>
<p><b>Note:</b> a:hover MUST come after a:link and a:visited in the CSS definition in order to be effective.</p>
<p><b>Note:</b> a:active MUST come after a:hover in the CSS definition in order to be effective.</p>
```

Styling a link with background-color property

This is a link

Note: a:hover MUST come after a:link and a:visited in the CSS definition in order to be effective.

Note: a:active MUST come after a:hover in the CSS definition in order to be effective.

Comments in HTML & CSS

- ```
  <!DOCTYPE html>
  <html>
  <head>
  <style>
  p {
    color: red; /* Set text color to red */
  }
  </style>
  </head>
  <body>
  <h2>My Heading</h2>
  <!-- These paragraphs will be red -->
  Hello World!
  This paragraph is styled with CSS.
  CSS comments are not shown in the output.
  </body>
  </html>
  ```
  ```

Border Color

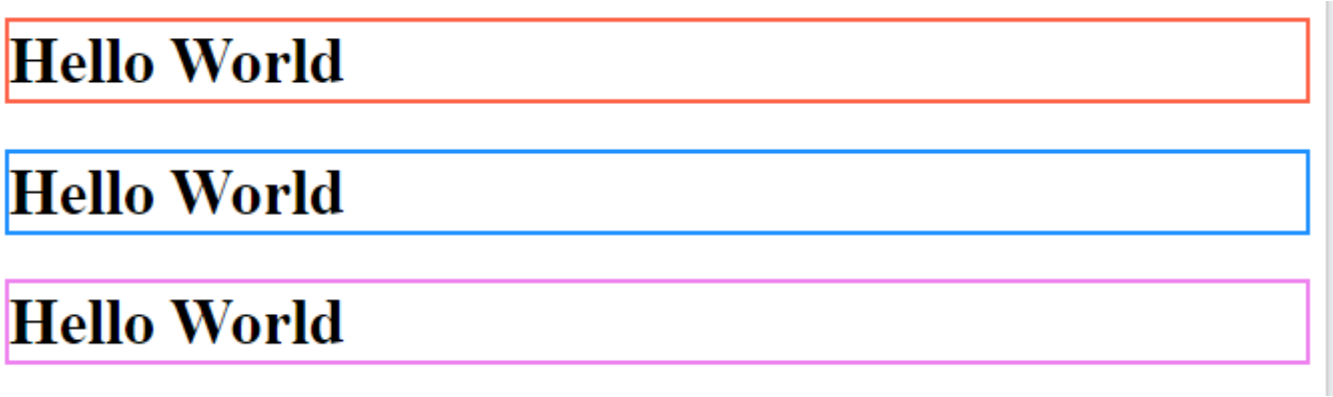
```
<!DOCTYPE html>
<html>
<body>

<h1 style="border: 2px solid Tomato;">Hello World</h1>

<h1 style="border: 2px solid DodgerBlue;">Hello World</h1>

<h1 style="border: 2px solid Violet;">Hello World</h1>

</body>
</html>
```



Round Corners

- The CSS border-radius property defines the radius of an element's corners.
- **Four values - border-radius: 15px 50px 30px 5px;** (first value applies to top-left corner, second value applies to top-right corner, third value applies to bottom-right corner, and fourth value applies to bottom-left corner)
- **Three values - border-radius: 15px 50px 30px;** (first value applies to top-left corner, second value applies to top-right and bottom-left corners, and third value applies to bottom-right corner)

- **Two values - border-radius: 15px 50px;** (first value applies to top-left and bottom-right corners, and the second value applies to top-right and bottom-left corners)
- **One value - border-radius: 15px;** (the value applies to all four corners, which are rounded equally

```css
#rcorners1 {
  border-radius: 25px;
  background: #73AD21;
  padding: 20px;
  width: 200px;
  height: 150px;
}

#rcorners2 {
  border-radius: 25px;
  border: 2px solid #73AD21;
  padding: 20px;
  width: 200px;
  height: 150px;
}

#rcorners3 {
  border-radius: 25px;
  background: url(paper.gif);
  background-position: left top;
  background-repeat: repeat;
  padding: 20px;
  width: 200px;
  height: 150px;
}
```



CSS Animation

- CSS allows animation of HTML elements without using JavaScript.
- **Properties**
  - @keyframes
  - animation-name
  - animation-duration
  - animation-delay
  - animation-iteration-count
  - animation-direction
  - animation-timing-function
  - animation-fill-mode
  - animation

CSS Animation

- An animation lets an element gradually **change from one style to another**.

- To use CSS animation, you must first specify some keyframes for the animation.

The @keyframes Rule

- When you specify CSS styles inside the **@keyframes rule, the animation will gradually change from the current style to the new style at certain times**.

- The **animation-duration property** defines how long an animation should take to complete. **If the animation-duration property is not specified, no animation will occur, because the default value is 0s (0 seconds)**.

```
<html>
<head>
<style>
div {
  width: 100px;
  height: 100px;
  background-color: yellow;
  animation-name: animation;
  animation-duration: 4s;
}
@keyframes animation {
  from {background-color: yellow;}
  to {background-color: green;}
}
</style>
</head>
<body>
<h1>CSS Animation</h1>
<div></div>
<p><b>Note:</b> When an animation is finished, it goes back to its original style.</p>
</body>
</html>
```

```
<html>
<head>
<style>
div {
  width: 100px;
  height: 100px;
  background-color: red;
  animation-name: example;
  animation-duration: 4s;
}
@keyframes example {
  0%   {background-color: red;}
  25%  {background-color: yellow;}
  50%  {background-color: blue;}
  100% {background-color: green;}
}
</style>
</head>
<body>
<h1>CSS Animation</h1>
<div></div>
<p><b>Note:</b> When an animation is finished, it goes back to its original style.</p>
</body>
</html>
```

Delay an Animation

- The animation-delay property specifies a **delay for the start of an animation.**

```
html>
<head>
<style>
div {
  width: 100px;
  height: 100px;
  background-color: red;
  position: relative;
  animation-name: example;
  animation-duration: 4s;
  animation-delay: 2s;
}
@keyframes example {
  0%   {background-color:red; left:0px; top:0px;}
  25%  {background-color:yellow; left:200px; top:0px;}
  50%  {background-color:blue; left:200px; top:200px;}
  75%  {background-color:green; left:0px; top:200px;}
  100% {background-color:red; left:0px; top:0px;}
}
</style>
</head>
<body>
<h1>CSS Animation</h1>
<p>The animation-delay property specifies a delay for the start of an animation. The following example has a 2 seconds delay before starting the animation:</p>

<div></div>

</body>
</html>
```

Iteration

- The animation-iteration-count property specifies the number of times an animation should run.

```
<html>
<head>
<style>
div {
  width: 100px;
  height: 100px;
  background-color: red;
  position: relative;
  animation-name: example;
  animation-duration: 4s;
  animation-iteration-count: 3/infinite;
}
@keyframes example {
  0%   {background-color:red; left:0px; top:0px;}
  25%  {background-color:yellow; left:200px; top:0px;}
  50%  {background-color:blue; left:200px; top:200px;}
  75%  {background-color:green; left:0px; top:200px;}
  100% {background-color:red; left:0px; top:0px;}
}
</style>
</head>
<body>
<h1>CSS Animation</h1>
<p>The animation-iteration-count property specifies the number of times an animation should run. The following example will run the animation 3 times before it stops:</p>
<div></div>
</body>
</html>
```

animation-direction

- The animation-direction property specifies whether an animation should be played forwards, backwards or in alternate cycles.

- The **animation-direction property** can have the following values:
  - normal - The animation is played as **normal (forwards).** This is default
  - reverse - The animation is played in **reverse direction (backwards)**
  - alternate - The animation is played **forwards first, then backwards**
  - alternate-reverse - The animation is played **backwards first, then forwards**

```
<html>
<head>
<style>
div {
  width: 100px;
  height: 100px;
  background-color: red;
  position: relative;
  animation-name: example;
  animation-duration: 4s;
  animation-direction: reverse;
}
@keyframes example {
  0%   {background-color:red; left:0px; top:0px;}
  25%  {background-color:yellow; left:200px; top:0px;}
  50%  {background-color:blue; left:200px; top:200px;}
  75%  {background-color:green; left:0px; top:200px;}
  100% {background-color:red; left:0px; top:0px;}
}
</style>
</head>
<body>

<h1>CSS Animation</h1>

<p>The animation-direction property specifies whether an animation should be played forwards, backwards or in alternate cycles. The following example will run the animation in reverse direction (backwards):</p>

<div></div>

</body>
</html>
```

animation-timing-function

The `animation-timing-function` property specifies the speed curve of the animation.

The animation-timing-function property can have the following values:

- `ease` - Specifies an animation with a slow start, then fast, then end slowly (this is default)
- `linear` - Specifies an animation with the same speed from start to end
- `ease-in` - Specifies an animation with a slow start
- `ease-out` - Specifies an animation with a slow end
- `ease-in-out` - Specifies an animation with a slow start and end
- `cubic-bezier(n,n,n,n)` - Lets you define your own values in a cubic-bezier function

Animation Shorthand Property

The example below uses six of the animation properties:

Example

```
div {
  animation-name: example;
  animation-duration: 5s;
  animation-timing-function: linear;
  animation-delay: 2s;
  animation-iteration-count: infinite;
  animation-direction: alternate;
}
```

Try it Yourself »

The same animation effect as above can be achieved by using the shorthand `animation` property:

Example

```
div {
  animation: example 5s linear 2s infinite alternate;
}
```

Try it Yourself »

Pagination



```
<!DOCTYPE html>
<html>
<head>
<style>
.pagination {
  display: inline-block;
}

.pagination a {
  color: black;
  float: left;
  padding: 8px 16px;
  text-decoration: none;
}
</style>
</head>
<body>

<h2>Simple Pagination</h2>

<div class="pagination">
  <a href="#">&amp;laquo;</a>
  <a href="#">1</a>
  <a href="#">2</a>
  <a href="#">3</a>
  <a href="#">4</a>
  <a href="#">5</a>
  <a href="#">6</a>
  <a href="#">&amp;raquo;</a>
</div>

</body>
</html>
```

```
<!DOCTYPE html>
<html>
<head>
<style>
.pagination {
  display: inline-block;
}

.pagination a {
  color: black;
  float: left;
  padding: 8px 16px;
  text-decoration: none;
}

.pagination a.active {
  background-color: black;
  color: white;
}

.pagination a:hover:not(.active) {background-color: yellow;}
</style>
</head>
<body>

<h2>Active and Hoverable Pagination</h2>

<p>Move the mouse over the numbers.</p>

<div class="pagination">
  <a href="#">&laquo;</a>
  <a href="#">1</a>
  <a class="active" href="#">2</a>
  <a href="#">3</a>
  <a href="#">4</a>
  <a href="#">5</a>
  <a href="#">6</a>
  <a href="#">&raquo;</a>
</div>

</body>
</html>
```

CSS scroll-behavior

Value	Description
auto	Allows a straight jump "scroll effect" between elements within the scrolling box. This is default
smooth	Allows a smooth animated "scroll effect" between elements within the scrolling box.
initial	Sets this property to its default value. Read about *initial*
inherit	Inherits this property from its parent element. Read about *inherit*

```
<html>
<head>
<style>
html {
  scroll-behavior: smooth;
}
#section1 {
  height: 600px;
  background-color: pink;
}
#section2 {
  height: 600px;
  background-color: yellow;
}
</style>
</head>
```

Property	Description
column-count	Specifies the number of columns an element should be divided into
column-fill	Specifies how to fill columns
column-gap	Specifies the gap between the columns
column-rule	A shorthand property for setting all the column-rule-* properties
column-rule-color	Specifies the color of the rule between columns
column-rule-style	Specifies the style of the rule between columns
column-rule-width	Specifies the width of the rule between columns
column-span	Specifies how many columns an element should span across
column-width	Specifies a suggested, optimal width for the columns
columns	A shorthand property for setting column-width and column-count

```
<style>
.newspaper {
  column-count: 3;
  column-gap: 40px;
  column-rule: 1px solid lightblue;
}
h2 {
  column-span: all;
}
</style>
```

Lorem Ipsum Dolor Sit Amet

Lorem ipsum dolor sit amet, consectetuer adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat. Ut wisi enim ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis nisl ut aliquip ex ea commodo consequat. Duis autem vel eum iriure dolor in hendrerit in vulputate velit esse molestie consequat, vel illum dolore eu feugiat nulla facilisis at vero eros et accumsan et iusto odio dignissim qui blandit praesent luptatum zzril delenit augue duis dolore te feugait nulla facilisi. Nam liber tempor cum soluta nobis eleifend option congue nihil imperdiet doming id quod mazim placerat facer possim assum.